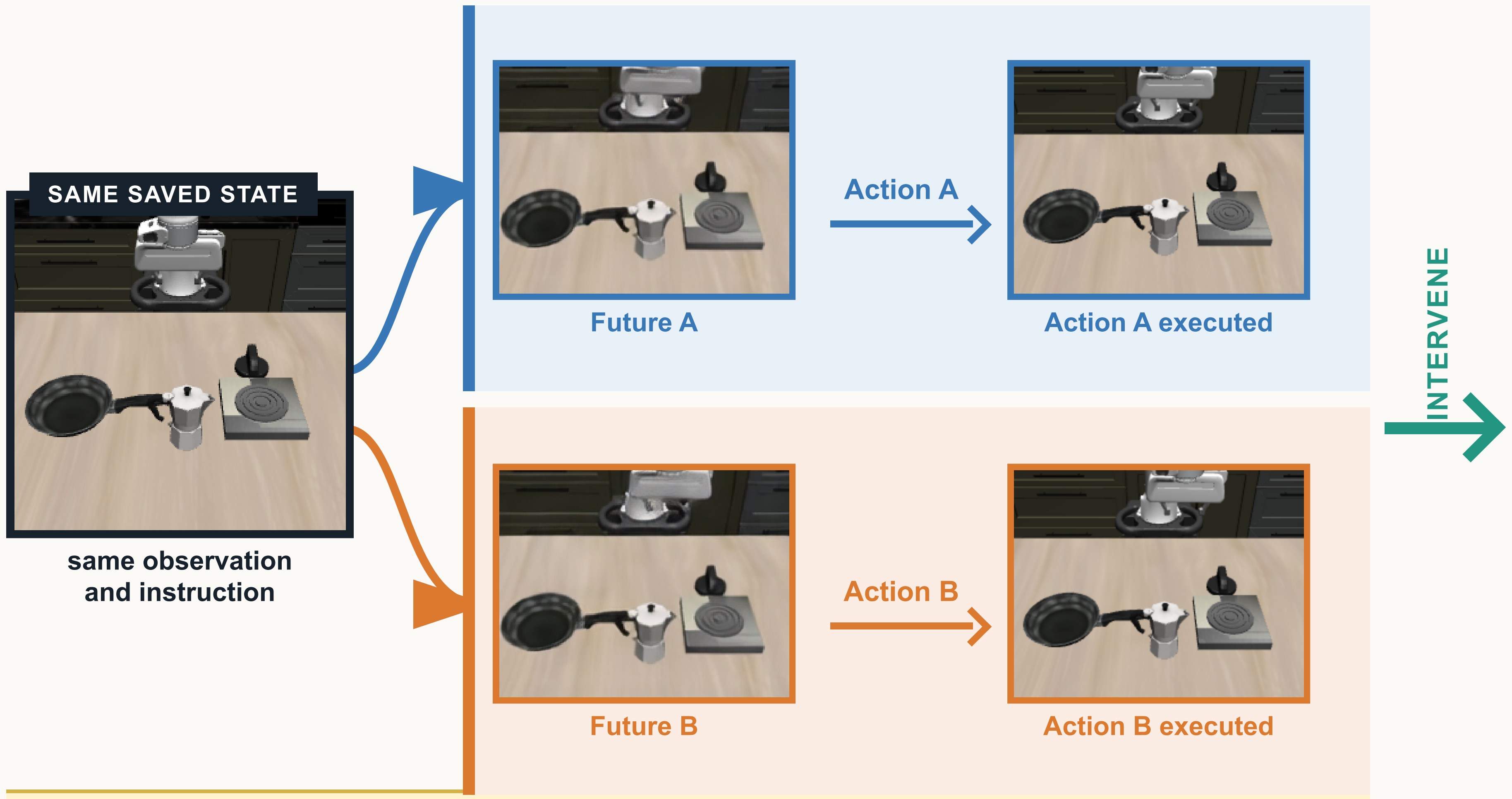
THE CAUSAL QUESTION

Does the generated future change the action?

A future and an action can be generated together without one causing the other. The test is to change *only the future* and ask whether the action follows.

WHAT NORMAL ROLLOUTS SHOW

One starting state, two different future–action pairs



Future A agrees with Action A; Future B agrees with Action B—but agreement alone is only correlation.

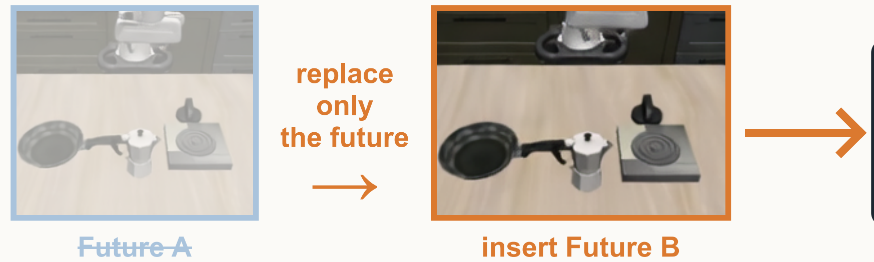
THE EXPERIMENT THAT SEPARATES CAUSE FROM CORRELATION

Rerun A, but insert Future B

If the future is incidental: the new action stays near A.
If its content controls action: the new action moves toward B.

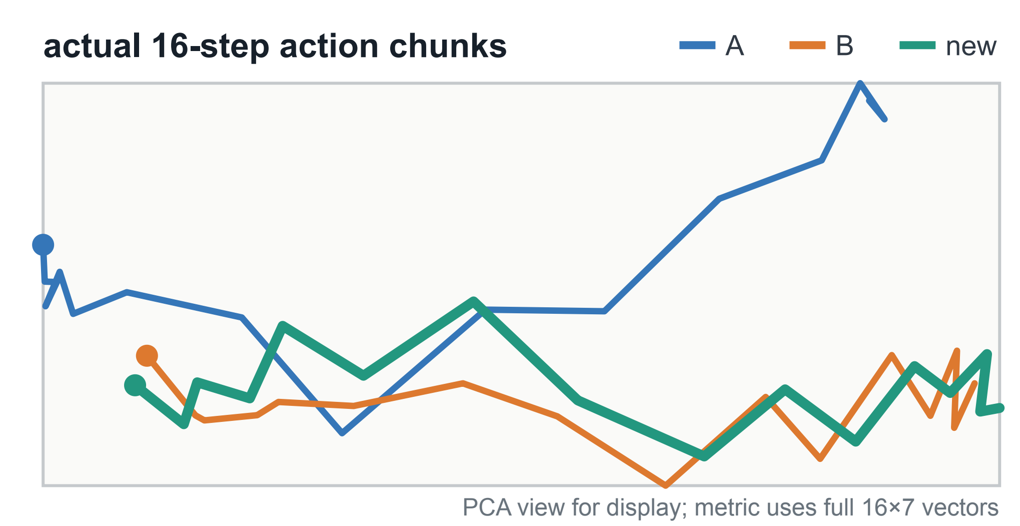
HELD FIXED FROM ROLLOUT A

same instruction "turn on the stove and put the moka pot on it"	**A's random draws** action noise + future noise	**same solver and schedule** 80 · 42.3 · 21.0 · 9.6 · 4



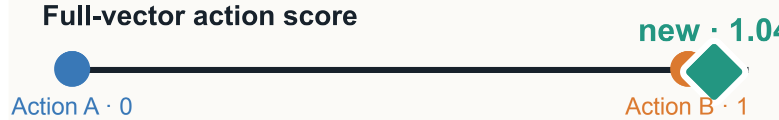
same policy computation

recompute A while repeatedly clamping only its future representation



Action B is withheld
It never enters the policy. Only after inference does it define the direction used for scoring.

Full-vector action score

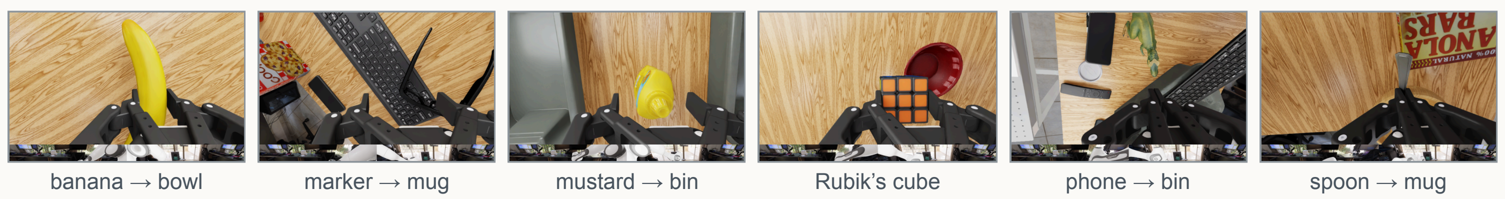


Because the new action moves toward B, Future B's content causally affects the action.

The example is illustrative; inference is population-level

2 models · 2 benchmarks

All prespecified states and all missing states are accounted for.



24 prespecified RoboLab states → **22** completed 22/22 positive

2 missing, disclosed and not replaced: their registered source videos ended before the required 33 frames. No intervention outcome entered retention.

- ✓ Six Cosmos 3 tasks; self, matched-Gaussian, and natural-future controls
- ✓ Task-balanced resampling preserves the conclusions
- ✓ Pairs selected from native success, separation, and replay validity only
- ✓ Cosmos Policy: 30 frozen LIBERO states/10 tasks; self, Gaussian, natural, and shuffled controls

What is actually replaced?

The overview stays plain-language; the intervention itself is tensor-level and repeated.

MODEL	TARGET	WHEN	NEVER WRITTEN
Cosmos Policy	Clean x_0 future latent: proprioception, wrist RGB, primary RGB at temporal slots 5–7	Initialization + every one of five solver evaluations; target reset after each evaluation. Conditional-only, so no CFG branch.	Action and value coordinates; current observation and instruction
Cosmos 3	Future-video latent frames 1–8. Noisy state uses $(1-\sigma)F^B + \sigma\varepsilon^A$; post-guidance future velocity is reset to the clean target.	Before conditional and unconditional forwards at all four UniPC calls; velocity reset after guidance each time	Current-video frame; action state and action velocity

Which visual content is sufficient?
Cosmos 3: disjoint robot/object pixel interventions at 10 states across six tasks. **Neither isolated factor recreates the whole-future effect.**

Which route carries the effect?
Self-future K/V replaces alternative-future K/V at all 36 layers × eight forwards, action-query rows only; other tokens and the future clamp stay fixed. **83–88% of Cosmos 3 steering is removed.**

Held-out LIBERO example: task02 · state10 · rollout seeds 313 and 347 · intervention noise seed 419 · actual 16×7 action chunks

This tests content-specific future conditioning. It does not show candidate comparison, task-success mediation, or task-level planning.